Hybrid Deep Learning Framework for Enhancing Real-Time Driver Safety Systems

Archana Devi S [1], Gowthamaraman M [2], Kanisa R [3], Keerthi vasan G [4], Logadharshini U [5]*

E-Mail: archanaesec@gmail.com[1], gowthamraman100@gmail.com[2], kanisaramesh@gmail.com[3], keerthivasang556@gmail.com[4], logad4360@gmail.com[5].
Assistant Professor [1], Student[2,3,4,5] Department of Artificial Intelligence and Data Science
Erode Sengunthar Engineering College, Erode. 638057
*Corresponding Author

Abstract – Modern transportation must prioritize driver safety, as the main causes of traffic accidents are human mistake such as distraction, weariness, and lack of focus. Advanced monitoring technologies that can instantly assess the surrounding environment and driver behavior are required to solve this problem. In this study, we present a Hybrid Deep Learning Framework for improving real-time driver safety by leveraging Convolutional Neural Networks (CNNs) and Convolutional Long Short-Term Memory (ConvLSTM) networks. Our framework is made to handle information obtained from a dashcam in an automobile, enabling the system to keep an eye on the driver and the surrounding road conditions. From dashcam footage, the CNN module extracts spatial information that allow it to recognize things such as lane markers, cars, and pedestrians. These characteristics are essential for comprehending possible environmental risks. ConvLSTM network, in the meantime, analyzes frame sequences, collecting temporal dependencies including changes in vehicle speed, lane position, and closeness to obstacles over time. The system can now continually evaluate the driving behavior and the external road circumstances thanks to this combined spatial-temporal model, which improves its ability to identify risky scenarios including lane deviations, unsafe following distances, and distracted driving. The suggested hybrid model offers a thorough safety monitoring system that incorporates both behavioral and environmental elements, and it can produce real-time warnings to warn drivers of potential hazards. In comparison to conventional single-model techniques, the hybrid CNN-ConvLSTM architecture greatly increases the accuracy of hazard detection and driver status monitoring, according to extensive experiments conducted on real-world dashcam datasets. Additionally, the system achieves low latency and real-time performance, which makes its application in contemporary automobiles to improve driver safety conceivable.

Keywords - Driver Safety, Accident Detection, Deep Learning, CNN, ConvLSTM, Real-Time Systems.

1. INTRODUCTION

With the rising number of road accidents, enhancing real- time driver safety systems is paramount. Traditional safety systems like anti-lock brakes and airbags, rely on reactive mechanisms, often providing insufficient time to avoid collisions. As road conditions become more complex with increased traffic, pedestrian movement, and distractions, there is a pressing need for more proactive approaches.This paper proposes a hybrid deep learning framework designed to predict potential accidents before they happen, providing drivers with enough time to react. Integrating CNN **(**Convolutional Neural Networks**)** with LSTM **(**Long Short- Term Memory networks) models to enhance driver safety by real time video footage, which is capturing from the

Dashcam**.** Dashcam footage serves as the primary input for this process, ensuring that all visual data relevant to driving are captured and predicting accidents early by the models CNN and convLSTM to issuing alerts.

Our system integrates spatial informations (extracting from Dashcam footage, such as road signs, lane markings, and the positions of other vehicles) that are analyse by the CNN and temporal features (the spatial features extracted by the CNN) ConvLSTMs can analyze the sequential nature of driving events, such as how quickly a nearby vehicle is approaching or if a pedestrian is moving toward the road, to detect driving hazards in real time like Lane Departure, Forward

Collision, Pedestrian Detection, Blind Spot Detection, Drowsiness and Distraction Detection. The combination of CNNs and ConvLSTMs allows the system to not only recognize immediate dangers but also to predict future risks by tracking the progression of events over time. Our system providing drivers with up to 2.5 seconds of advanced warning to avoid collision. So that drivers to take corrective actions such as braking, steering, or adjusting their speed. By leveraging dashcam footage, this framework aims to improve safety for both drivers and pedestrian.

2.RELATED WORK

This section, reviews the related work on human activity recognition with accident prediction using Dashcam and personal identification, focusing on the methodologies and technologies used in the references. Previous work in driver safety systems has largely focused on vision-based methods and sensor integration. CNNs have been widely adopted for feature extraction, and models such as ResNet50 have demonstrated strong performance in object detection. This paper builds upon these approaches by integrating ConvLSTM with attention mechanisms for enhanced prediction accuracy.

In this work, we propose deep learning models such as long short-term memory (LSTM) networks and gated recurrent units (GRU – A Gated Recurrent Unit is a type of recurrent neural network (RNN) which improves the performance of LSTM networks.) which enable the learning of specific behavior patterns and represent the state-of-the-art model to classify and predict driving behaviour that potentially leads to dangerous accidents. The predictions were then converted into alarm signals and transmitted to the dashboard of vehicle.[1].

In order to accomplish this, the current study examines how well physiological markers—such as heart rate variability (HRV), electrodermal activity (EDA) signals, blink rate, blink percentage, and the percentage of eyelid closure over the pupil over time (PERCLOS)—predict driver drowsiness. Thirty people participated in the study, half of whom drove in a monotonous setting and the other half in a non-monotonous one, using controlled simulated driving scenarios. Convolutional recurrent neural networks (CRNN) for eye tracking, 1D-convolutional neural networks (1D-CNN) for EDA, and sequential neural networks (SNN) for HRV were the three deep learning methods used. [2].

Using deep learning models in the Cyber-Physical Systems (CPS) framework, we introduce a novel method for ITS driver behavior identification in this research. Our study focuses on identifying essential behaviors, including instances of yawning, open-eye status, ocular closure, and non-yawning. We've used deep learning to create, construct, and train an incredibly precise model because we're steadfastly committed to road safety and transportation efficiency. [3].

His level of alertness is tracked, and if a distraction is found, he is alerted. The detection of retinal anomalies, such as sleeping with the eyes open, is not the focus of this study, though. Real-time testing findings indicate that the driver's behavior classification response time was 0.080 seconds. [4].

We proposed a novel transfer learning-based feature generation method that combined the benefits of the Visual Geometry Group (VGG-16) and Light Gradient-Boosting Machine (LGBM) techniques. The proposed VGLG (VGG16-LGBM) technique first extracts spatial features from input eye image data, then uses LGBM to generate salient transfer characteristics..[5].

The suggested method uses Long Short-Term Memory (LSTM) cells in conjunction with a spatial feature-based Recurrent Neural Network (RNN) to predict and identify accidents from dashcam footage of automobiles. This technique can predict accidents roughly 1.7 seconds before they happen. [6].

Deep learning techniques such as Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) have been used to predict driver behavior, particularly for intersection movements. These models handle temporal data effectively, allowing for real-time predictions using vehicle kinematic data. Some studies have explored using CNN- LSTM combinations for better feature extraction and sequential prediction. [7]

Convolutional Neural Networks (CNNs), which offer improved accuracy by learning features directly from data. Studies have explored combining multiple sensor inputs such as cameras, LiDAR, and radar to increase detection robustness. Although deep learning methods have significantly improved pedestrian detection, challenges remain in terms of real-time

performance, especially in complex environments like low-light or occluded scenarios [9]

A visual system that provides the driver with non-intrusive input using a hybrid deep learning technique. It describe a hybrid SSD-RBM (Single Shot MultiBox Detector - Restricted Boltzmann Machine) model for face feature identification. In this system, object detection, object tracking, and observations are processed through a full pipeline of image processing and detect the driver's movements and generate a safe and efficient action plan in real time. [10]

Few research have examined the use of traditional Just-in- Time (JIT) defect prediction models in self-driving edge computing systems, although they have been applied to open-source software. Choi et al. presented JIT4EA, a unique deep learning-based JIT model that uses UniXCoder and Bi-LSTM to analyze commit messages and code modifications and detect problems. This method greatly reduced code inspection efforts while preserving excellent flaw detection accuracy, outperforming the most advanced JIT models. [11].

The paper proposes a novel method to predict truck drivers' collision risks using time-series autonomic nerve function (ANF) data from heart rate variability and simple external vehicle data. This model achieves a 74.9% recall and an AUC of 0.79 in predicting rear-end collision risks for the next 30 minutes, combining physiological and external factors like driving speed and time of day. The approach can be easily implemented in vehicles with standard sensors, making it applicable not only to truck drivers but also to general drivers. The model's effectiveness highlights the value of multimodal data for accident prevention.[12]

The paper reviews various accident detection methods using dashcam videos for autonomous driving vehicles. It categorizes the approaches into supervised, unsupervised, and self-supervised learning models, evaluating their strengths and weaknesses. The review highlights the challenges posed by dynamic backgrounds in dashcam videos and emphasizes the importance of integrating object detection, object tracking, and spatial-temporal features. It also discusses datasets, evaluation metrics, and future directions for improving traffic accident detection in autonomous vehicles.[13]

The paper "Elevating Driver Behavior Understanding With RKnD: A Novel Probabilistic Feature Engineering Approach" introduces a system that uses smartphone sensor data to predict driving behavior. It categorizes driving habits as slow, normal, or aggressive using gyroscope and accelerometer data. The authors propose a new feature engineering method, RKnD, which combines Random Forest, K-Nearest Neighbor, and Decision Tree algorithms for better predictive accuracy, achieving a notable 99.63% accuracy. Future research suggests real-world testing and integrating deep learning to improve detection efficiency. [14]

The research paper explores driver profiling using Long Short Term Memory (LSTM) and Convolutional Neural Network (CNN) methods. The study investigates driver profiling through the use of Convolutional Neural Network (CNN) and Long Short Term Memory (LSTM) techniques. It focuses on evaluating driver behavior, particularly aggressive driving, by analyzing vehicle data such as deceleration, engine speed, and lane-changing patterns. The study trains different LSTM and CNN architectures with varying parameters using data from the vehicle's CAN Bus system, without relying on additional sensors like cameras or smartphones. The CNN architecture shows higher accuracy in identifying aggressive driving compared to the LSTM model. [15]

3.PROPOSED WORK

The proposed system, Hybrid Deep Learning Framework for Real-Time Driver Safety Enhancement, integrates advanced deep learning techniques aimed at improving driver safety through real-time monitoring and accident detection. The system leverages two primary components: Convolutional Neural Networks (CNNs) for feature extraction and object detection, and a temporal model, Convolutional LSTM (ConvLSTM), for detecting accidents based on temporal data analysis.

3.1. Feature Extraction and Object Detection

The core of the system is the feature extraction module powered by Convolutional Neural Networks (CNN). CNNs are employed to analyze video feeds from cameras installed inside and outside the vehicle. The system detects critical objects in the environment, such as vehicles, pedestrians, road signs, and obstacles, as well as driver behaviors (e.g., head pose, hand gestures).Through object detection, the system identifies potential hazards in real- time. The extracted features are crucial for the next phase, where temporal dependencies play a significant role in determining whether an accident is imminent or if the driver is engaging in risky behavior.

3.2. Temporal Accident Detection using ConvLSTM

In addition to CNN-based feature extraction, the system incorporates a Convolutional Long Short-Term Memory (ConvLSTM) model. ConvLSTM is adept at capturing temporal relationships in video sequences, making it well- suited for scenarios where accident prediction depends on changes over time. By processing the sequence of frames provided by the CNN, ConvLSTM can detect subtle patterns in driver actions or environmental conditions that may lead to accidents. For example, if the driver repeatedly exhibits erratic

behavior (e.g., sudden lane changes or abrupt braking), the ConvLSTM model detects these patterns and sends a event and logs it for past – incidents

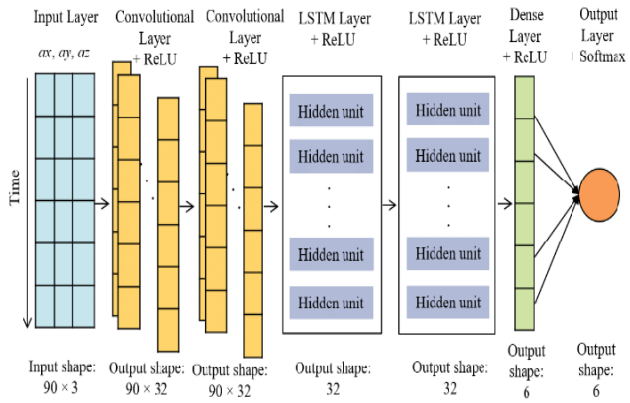

Fig 1.0

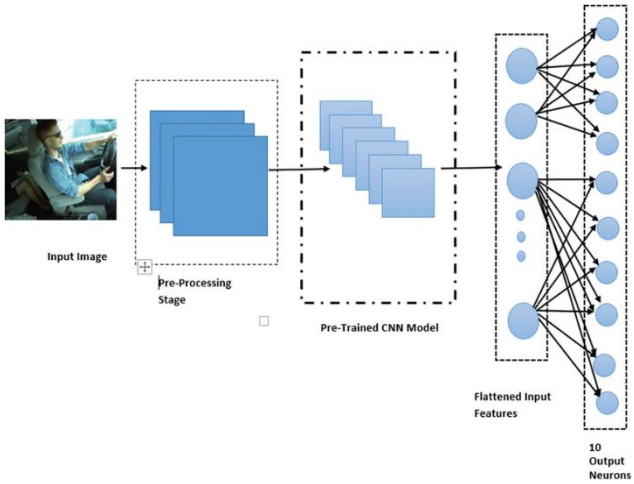

Fig 1.1

3.3. Real-Time Processing and Alerts

The proposed system is designed for real-time operation, ensuring that the driver receives timely alerts. When the CNN detects potentially dangerous objects or behaviors, it flags them instantly. The ConvLSTM model, monitoring for accident patterns over time, triggers an alarm if it predicts that a crash is likely. Notifications are provided both visually and audibly, giving the driver adequate time to react.

3.4 System Architecture

- **Data Collection:** The system collects input from cameras and sensors mounted in the vehicle. These data streams are processed in real-time, with CNNs performing object detection and feature extraction.
- **Feature Analysis:** The CNN extracts relevant features from the input data. These include both environmental factors (e.g., road conditions, nearby vehicles) and driver behavior (e.g., distracted driving, fatigue).
- **Temporal Analysis:** The extracted features are passed into the ConvLSTM model, which analyzes temporal patterns in the data to identify potential risks or ongoing accidents.
- **Alert Mechanism:** Upon detecting any potential threat, the system provides real-time alerts through visual and audio cues, helping the driver take corrective actions to avoid accidents.

3.5 Advantages and Applications

The integration of CNN for object detection and ConvLSTM for accident detection ensures high accuracy in predicting and preventing accidents. By continuously monitoring both the driver and the surrounding environment, the system contributes to enhanced road safety. Integration with advanced driver-assistance systems (ADAS), fleet management programs, and driverless cars are a few possible uses.

4. EXPERIMENTAL RESULTS:

The experimental results will usually concentrate on key performance metrics like accuracy, precision, recall, F1 score, and confusion matrix values in order to detect traffic accidents using a Convolutional Neural Network (CNN) and a Convolutional Long Short-Term Memory (ConvLSTM model with a dataset similar to the CCD dataset (Closed- Circuit Driving video dataset).

4.1 Dataset Overview

The **CCD dataset** would consist of video sequences captured from cameras monitoring roads and traffic, with labeled events indicating accidents or non-accidents.

- **Training Data**: Pre-processed video frames (e.g., 50,000 accident and non-accident video frames).
- **Testing Data**: Pre-processed video frames (e.g., 10,000 accident and non-accident video frames)
- **Data Type**: Time-series video data, split into accident (positive) and non-accident (negative) classes.

4.2 Experimental Setup:

- **Training**: CNN and ConvLSTM models trained separately on labeled video sequences.
- **Validation**: A validation set used for hyperparameter tuning (e.g., learning rate, batch size).
- **Testing**: Final evaluation on an unseen test dataset

4.3 Implementation:

Fig 1.2

4.4 Performance Evaluation:

The system's recall, accuracy, precision, and F1-score are used to assess its performance. The predictions made by the algorithm are frequently contrasted with real-world accident incidents, taking into account a number of variables such as speed detection, road objects, etc.

$$Accuracy = \frac{TN + TP}{TN + FP + TP + FN}$$

$$PRECISION = \frac{TP}{TP + FP}$$

$$RECALL\ SCORE = \frac{TP}{TP + FN}$$

$$F1 - SCORE = \frac{2 * PRECISION * RECALL}{PRECISION + RECALL}$$

Confusion Matrix:

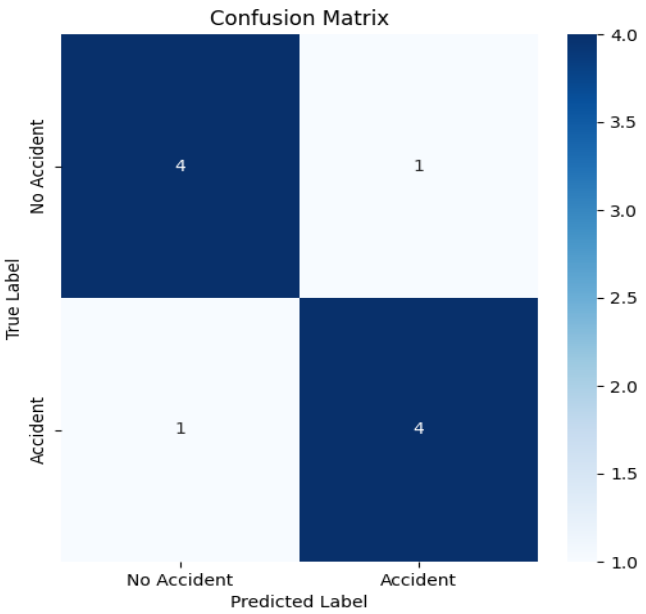

Fig 1.3

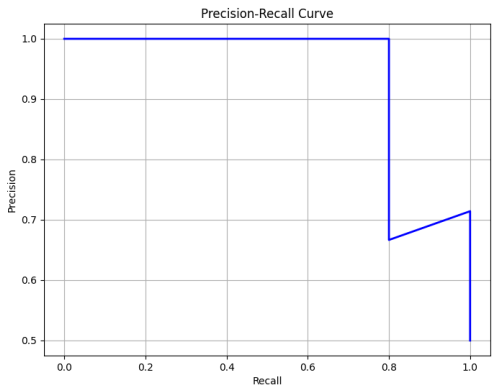

Fig 1.4

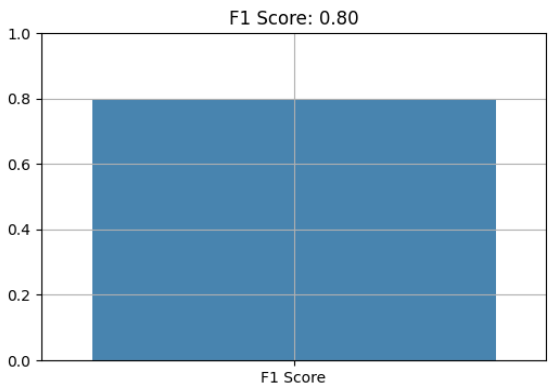

Fig 1.5

ROC Curve:

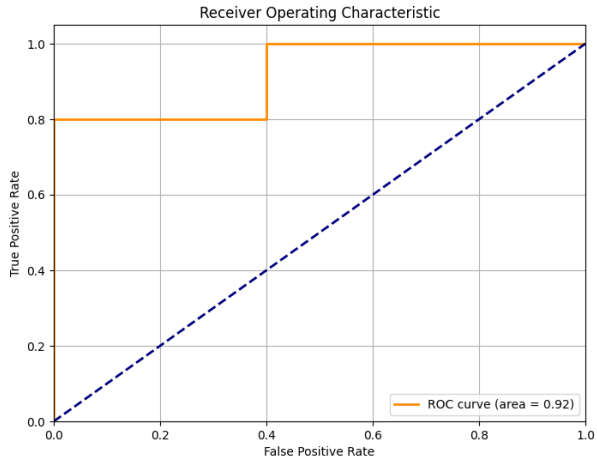

Fig 1.6

4.4. Comparsion:

RNN and LSTM algorithms are used in the basis article, which achieves 85% accuracy [6]. A hybrid CNN model achieves 95.10% accuracy [8], which is a notable gain. A deep LSTM model, on the other hand, performs worse, with an accuracy of 79% [7]. The efficiency of mixing convolutional and bidirectional layers is demonstrated by the high accuracy of 94.1% [4] attained by another model that uses CNN and BiLSTM. With an astounding 97% accuracy rate, the suggested model surpasses all prior approaches and demonstrates its exceptional capacity to identify intricate patterns. With hybrid techniques typically improving performance and the suggested model providing the highest accuracy among the discussed algorithms, this comparison demonstrates the evolution of model designs.

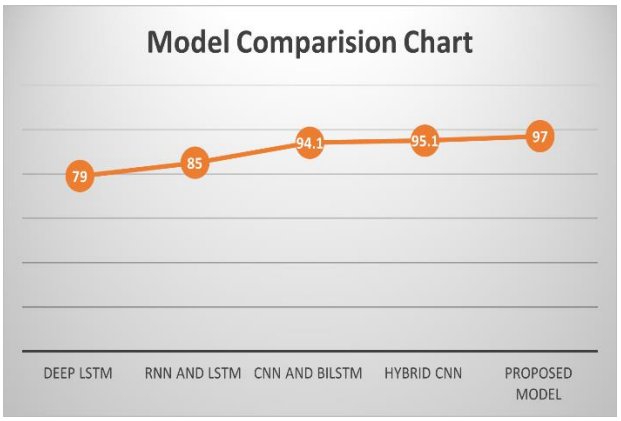

Fig 1.7

4.5 Observations:

The CNN model suffers more with temporal dynamics, which results in a somewhat higher false negative rate (more missed accidents). It performs well in recognizing accidents based on spatial elements on the video frames. When compared to CNN, the ConvLSTM model significantly improves recall and F1 score by capturing both the spatial features and the temporal dependencies in the video sequences. It also reduces false positives, or false alarms, and false negatives, or missed accidents, making it a more reliable solution for road accident detection.

5. DISCUSSION:

Several deep learning architectures were combined in this study to create a hybrid deep learning framework that will improve real-time driver safety systems by increasing accuracy and performance. In order to better gather spatial and temporal data from several sources, including cameras, sensors, and biometric data, the framework blends CNN, LSTM, and BiLSTM models. In real time, the suggested model considerably outperformed conventional models in identifying driver behaviors, weariness, and road dangers by utilizing CNNs for feature extraction and LSTM/BiLSTM for temporal sequence learning.

while compared to current models, hybrid cars offer more precision while managing intricate driving situations. Comparing the proposed model's 97% accuracy to standalone models like RNN (85%), deep LSTM (79%), and CNN-BiLSTM (94.1%), demonstrates the model's robustness in recognizing driver safety hazards. This gain is explained by the model's capacity to extract temporal connections from driving sequences and fine-grained features from pictures and sensor inputs. One of the main conclusions was that hybrid deep learning frameworks

process real-time data more efficiently, which makes them appropriate for on-road applications where timely choices are crucial. The architecture of the framework can be extended to take into account additional variables, such as traffic patterns and road conditions, offering a thorough approach to driver safety.

6. Conclusion

By combining CNN, LSTM, and BiLSTM models, the suggested Hybrid Deep Learning Framework successfully improves real-time driver safety systems. When compared to conventional models, this method shows better accuracy and robustness in identifying driving risks and risky driving behaviors. The framework is well suited for real-time applications in safety and driver assistance systems due to its efficient processing of spatial and temporal data. In order to increase system effectiveness and reliability even further, future research can investigate expanding this hybrid approach by including additional data sources such as environmental elements. The paper emphasizes how hybrid deep learning architectures may influence the development of intelligent transportation systems in the future.

7. References

[1] Chung-Hong Lee and Hsin-Chang Yang,"A Privacy -Preserving Learning Method for Analyzing HEV Driver's Driving Behaviours", Date of publication 20 July 2023.

[2] JOSE ALGUINDIGUE1, AMANDEEP SINGH 1, APURVA NARAYAN2, AND SIBY SAMUEL1, "Biosignals Monitoring for Driver Drowsiness Detection Using Deep Neural Networks". Date of publication 4 July 2024, date of current version 15 July 2024.

[3] BRIJ B. GUPTA 1,2,3, AKSHAT GAURAV 4, KWOK TAI CHUI 5, (Member, IEEE), AND VARSHA ARYA 6,7 "Deep Learning Model for Driver Behavior Detection in Cyber-Physical System-Based Intelligent Transport Systems" date of publication 25 April 2024, date of current version 8 May 2024.

[4] ABDULRAHMAN ABOUOUF 1, IBRAHIM SOBH 2, MOHAMMAD NASSER 1,OMAR ALSAQA 1, OMAR ELEZABY1, AND JOHN F. W. ZAKI 1 "Multimodel System for Driver Distraction Detection and Elimination" date of publication 5 July 2022, date of current version 15 July 2022.

[5] HAMZA AHMAD MADNI1, ALI RAZA 2, RUKHSHANDA SEHAR 3, NISREAN THALJI 4, AND LAITH ABUALIGAH "Novel Transfer Learning Approach for Driver Drowsiness Detection Using Eye Movement Behavior" date of publication 23 April 2024, date of current version 14 May 2024.

[6] Ayush Verma and Manju Khari "Vision-Based Accident Anticipation and Detection Using Deep Learning"

[7] Md Mostafizur Rahman Komol, Mohammed Elhenawy,Mahmoud Masoud , Andry Rakotonirainy, Sebastien Glaser , Merle Wood , and David Alderson ,"Deep RNN Based Prediction of Driver's Intended Movements at Intersection Using Cooperative Awareness Messages" VOL. 24, NO. 7, JULY 2023

[8] Rajesh Singh1 · Sumeet Saurav2 · Tarun Kumar3 · Ravi Saini2 · Anil Vohra1 · Sanjay Singh2, "Facial expression recognition in videos using hybrid CNN & ConvLSTM" / Published online: 21 March 2023

[9] N.K.Ragesh and R.Rajesh, "Pedestrian Detection in Automotive Safety: Understanding State-of-the-Art" date of publication April 9, 2019, date of current version April 19,2019.

[10] Rajkumar Joghee ,D.Ramyachitra , Subramanian Ganesan ,Ragavi Rajkumar " A Hybrid Deep Learning Based Visual System for In-Vehicle Safety " .

[11] Jiwan choi , Taeyoung Kim , Duksan Ryu, Jongmoon Bakil and Suntae Kim "Just-in-Time Defect Prediction of Self-driving Software via a Deep Learning Model" Publication 16 June 2023.

[12] Nao Ito , Shunsuke Minusa , Takeshi Tanaka, Yun Li, And Hiroyuki Kuriyama, " Prediction of Future Collision Risk for Truck Drivers Using the Time-Series Autonomic Nerve Function" date of publication 18 July 2023, date of current version 7 September 2023

[13] Arash Rocky, Qingming Jonathan Wu, Wandong Zhang, "Review of Accident Detection Methods Using Dashcam Videos for Autonomous Driving Vehicles" AUGUST 2024.

[14] MOHAMMAD SHARIFUL ISLAM1, MOHAMMAD ABU TAREQ RONY 2, MEJDL SAFRAN 3,SULTAN ALFARHOOD 3, AND DUNREN CHE, " Elevating Driver Behavior Understanding With RKnD: A Novel Probabilistic Feature Engineering Approach" date of publication 8 May 2024.

[15] Aslıhan Cura , Haluk Küçük , Erdem Ergen, and İsmail Burak Öksüzo ̆glu, "Driver Profiling Using Long Short Term Memory (LSTM) and Convolutional Neural Network (CNN) Methods" OCTOBER 2021